Model Answers (Sample Paper 2025)

Level 2

Q1 (a) f_s (sampling frequency) $= \dfrac{10{,}000 \text{ bits/sec}}{10 \text{ bits/sample}}$

$= 1000$ samples/sec

$= 1$ kHz

1024 levels $= 2^{10}$ i.e 10 bits

(i) The frequencies of $x(t)$ are: f_1 & f_2

$x(t) = 5\cos(600\pi t) + 4\cos(1800\pi t)$

$f_1 = 300$ Hz $< \frac{f_s}{2}$; $f_2 = 900$ Hz $> \frac{f_s}{2}$

(not affected)

$\therefore$ It is affected by aliasing

$\therefore f_2' = 900 - k \cdot f_s = 900 - 1 \cdot 1000$

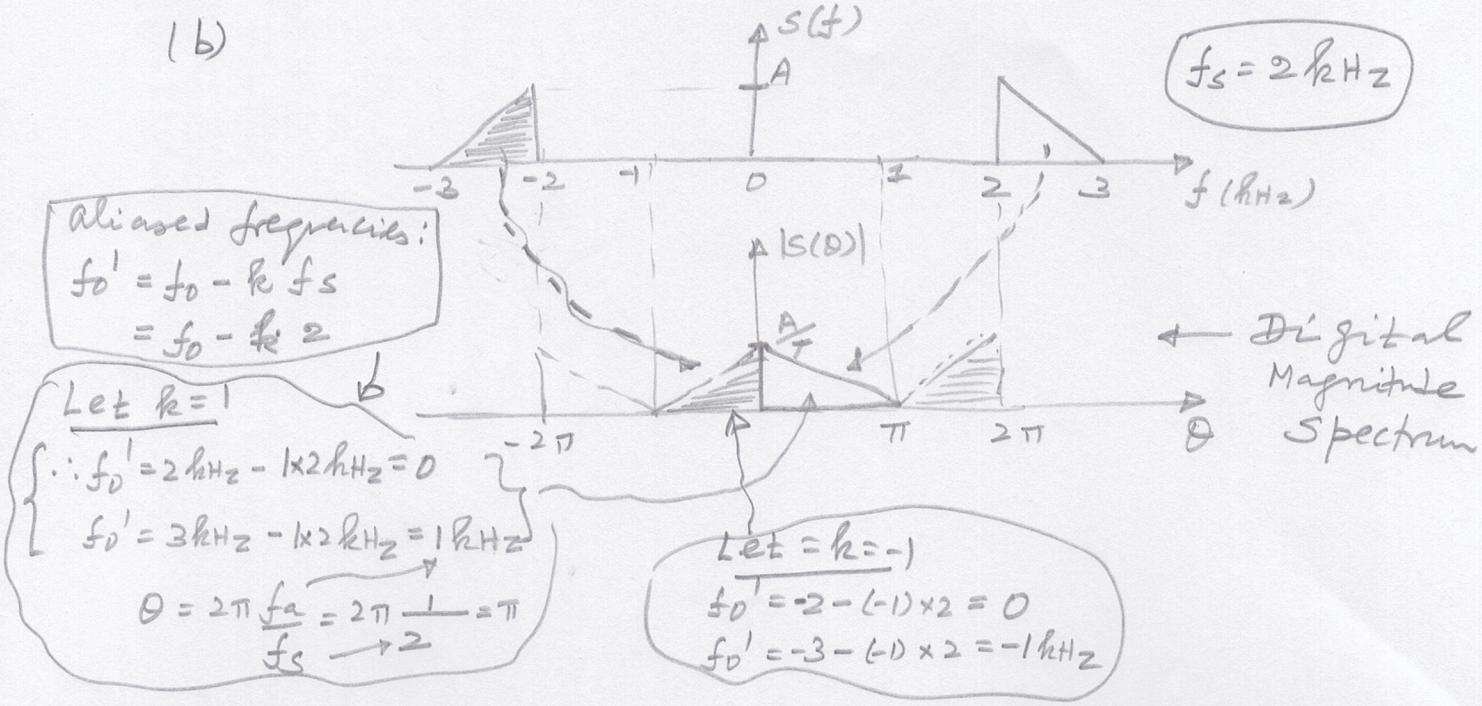
$= -100$ Hz

$\therefore f_1 = 300$ Hz ; $f_2 = -100$ Hz

(ii) $x[n] = 5\cos\left(2\pi \frac{300}{1000} n\right) + 4\cos\left(2\pi \frac{(-100)}{1000} n\right)$

$$x[n] = 5\cos\left(\frac{3\pi}{5} n\right) + 4\cos\left(\frac{\pi}{5} n\right)$$

(b)



Aliased frequencies:

$f_0' = f_0 - k f_s$

$= f_0 - k\,2$

Let $k = 1$

$\therefore f_0' = 2\text{kHz} - 1\times 2\text{kHz} = 0$

$f_0' = 3\text{kHz} - 1\times 2\text{kHz} = 1\text{kHz}$

$\Theta = 2\pi \frac{f_a}{f_s} = 2\pi \frac{1}{2} = \pi$

Let $k = -1$

$f_0' = -2 - (-1)\times 2 = 0$

$f_0' = -3 - (-1)\times 2 = -1$ kHz

(c) Peak to Peak amplitude = 10 Volts

(i) Quantisation interval or resolution $\Delta V = \frac{10V}{2^B - 1}$

$$\Delta V = \frac{10V}{2^B - 1} \approx \frac{10 \text{ Volts}}{2^B} \quad (\text{if } B \text{ is Large})$$

$$\therefore \Delta V = \frac{10}{2^8} = 39.062 \text{ mV}$$

8-bit A/D converter

(ii) Maximum quantisation error $= \frac{\Delta V}{2} = 19.53 \text{ mV}$

Q2 (a) (i) $H(z) = \frac{k_0}{(1 - b_1 z^{-1})^2} \Rightarrow \therefore H(\theta) = \frac{k_0}{(1 - b_1 e^{-j\theta})^2}$

$$|H(\theta)| = 1 \Rightarrow 1 = \frac{k_0}{(1 - b_1 \cdot 1)^2} \quad (\theta = 0)$$

$$\boxed{\therefore k_0 = (1 - b_1)^2}$$

(ii) $|H(\theta)|^2 = \frac{k_0}{(1 - b_1 e^{-j\theta})^2} \cdot \frac{k_0}{(1 - b_1 e^{j\theta})^2}$ $\left\{|H(\theta)|^2 = H(\theta) \cdot H^*(\theta)\right\}$

$$|H(\theta)|^2 = \frac{k_0^2}{(1 + b_1^2 - 2 b_1 \cos\theta)^2}$$

$$\underbrace{|H(\theta)|^2_{\theta = \pi/4}}_{\text{given}} = \frac{1}{2} = \frac{k_0^2}{\left(1 + b_1^2 - 2 b_1 \frac{1}{\sqrt{2}}\right)^2} \Rightarrow \left(1 + b_1^2 - b_1\sqrt{2}\right)^2 = 2 k_0^2$$

$$\Downarrow$$

$$\left(1 + b_1^2 - b_1\sqrt{2}\right) = \pm\sqrt{2}\, k_0$$

$$\underbrace{1 + b_1^2 - b_1\sqrt{2} = \pm\sqrt{2}(1 - b_1)^2}$$

$$\Downarrow$$

$b_1 = 0.32 \therefore k_0 = (1 - 0.32)^2 = 0.46$

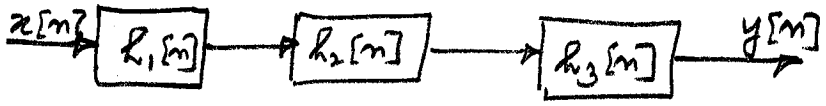
or

$b_1 = 3.09, \therefore k_0 = 4.37$

(iii) $H(z) = \alpha\left[\frac{1}{1 - \alpha z^{-1}} + \frac{z^{-1}}{1 - \alpha z^{-1}}\right]$

$$h[n] = \alpha\left[(\alpha)^n u[n] + (\alpha)^{n-1} u[n-1]\right] = \alpha^{n+1} u[n] + \alpha^n u[n-1]$$

$$\boxed{h[n] = \alpha^{n+1} u[n] + \alpha^n u[n-1]}$$

(b) 

$$h_c[n] = h_1[n] * h_2[n] * h_3[n]$$

$$h_c[n] = \{\delta[n] - \delta[n-1]\} * \{\delta[n] - \delta[n-1]\} * u[n]$$

z-domain solution:

z-transform:

$$\therefore H_c[z] = (1 - z^{-1}) \cdot (1 - z^{-1}) \cdot \frac{1}{1 - z^{-1}}$$

$$\therefore H_c(z) = 1 - z^{-1}$$

$$\therefore h_c[n] = \delta[n] - \delta[n-1]$$

OR

Time domain solution:

$$h_c[n] = \{\delta[n] - \delta[n-1]\} * \{\delta[n] - \delta[n-1]\} * u[n]$$

$$= \{\delta[n] - \delta[n-1]\} * \{\underbrace{u[n] - u[n-1]}_{=\delta[n]}\}$$

$$= \{\delta[n] - \delta[n-1]\} * \delta[n]$$

$$h_c[n] = \delta[n] - \delta[n-1]$$

Q3 $\theta_0 = 2\pi \frac{60}{300} = \frac{2\pi}{5}$; $\Delta\theta = 2\pi \frac{5}{300} = \frac{\pi}{30} = 0.1047$

$$r = 1 - 0.5\Delta\theta = 1 - 0.5 \times \frac{\pi}{30} = 1 - \frac{\pi}{60} = 0.9476$$

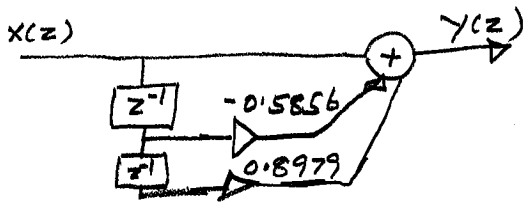
The transfer function of the notch filter

$$H(z) = \frac{(z - re^{j\frac{2\pi}{5}})(z - re^{-j\frac{2\pi}{5}})}{z^2}$$

$$= \frac{(z^2 - 2r\cos\frac{2\pi}{5} z + r^2)}{z^2}$$

$$H(z) = 1 - 2r\cos\frac{2\pi}{5} z^{-1} + r^2 z^{-2}$$

$$H(z) = 1 - 0.5856\, z^{-1} + 0.8979\, z^{-2}$$



(b) $H(z) = \dfrac{(z - 1 \cdot e^{j\frac{3\pi}{4}})(z - 1 \cdot e^{-j\frac{3\pi}{4}})}{(z + \frac{1}{2}j)(z - \frac{1}{2}j)} = \dfrac{z^2 - 2\cos\frac{3\pi}{4} z + 1}{(z^2 + \frac{1}{4})}$

$$\boxed{H(z) = \frac{z^2 + \sqrt{2}z + 1}{z^2 + \frac{1}{4}} = \frac{1 + \sqrt{2}z^{-1} + z^{-2}}{1 + \frac{1}{4}z^{-2}}}$$

(c) $h[n] = [1, 2, 3]$

$h[n] = [1, 2, 3, 0]$ ← $N = 4$

DFT : $H[k] = \sum_{n=0}^{3} h[n] e^{-jn\frac{2\pi}{4}k}$

$$H[k] = 1 + 2e^{-j\pi k} + 3e^{-j\frac{3\pi}{2}k} + 0$$

$$\therefore H[k] = 1 + 2e^{-j\pi k} + 3e^{-j\frac{3\pi}{2}k}$$

where $k = 0, 1, 2, 3$

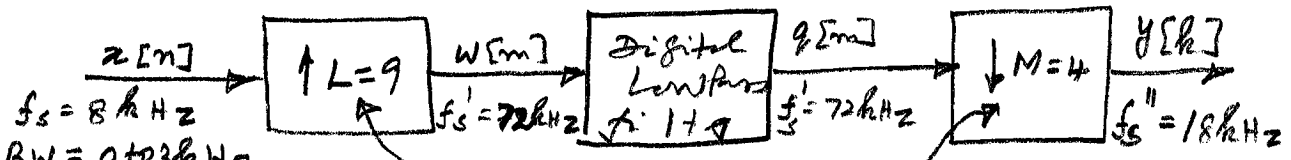
DC gain of the filter is $H[0]$

When $k = 0$, $H[0] = 1 + 2 + 3 = 6$

DC gain

Q4

(a)



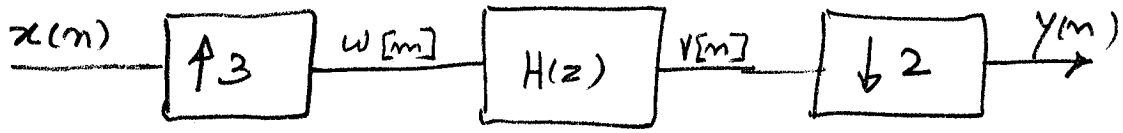
$\frac{L}{M} = \frac{18}{8} = \frac{2 \times 9}{2 \times 4}$

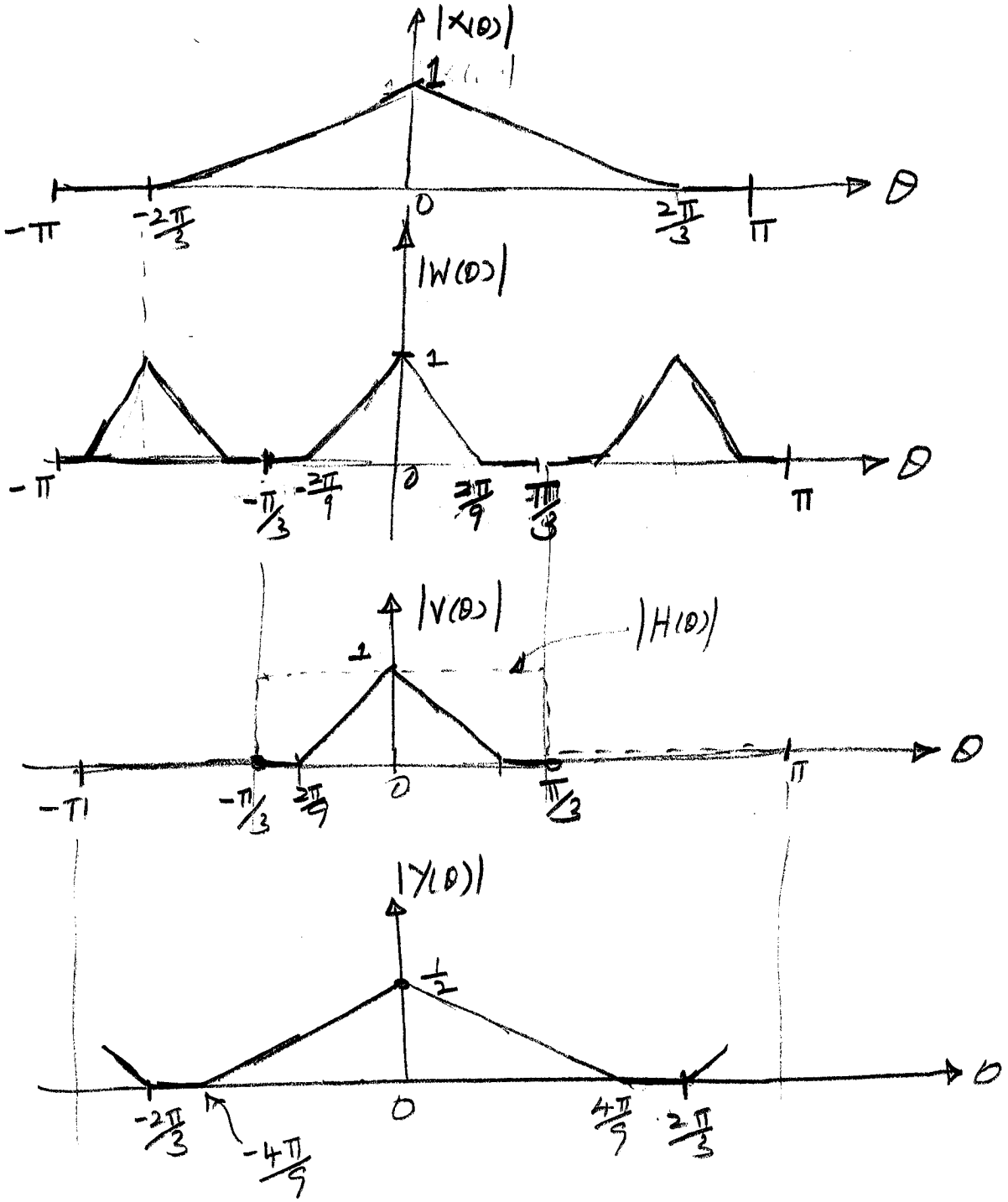
$f_s'' = \frac{9}{4} f_s$
$= \frac{9}{4} \times 8kHz$
$= 18kHz$

Using an IIR filter in this case has an obvious shortcoming, as we have to calculated every output (since previous outputs are required to compute M^{th} output) and then drop the outputs except M^{th} outputs. Since Previous outputs are not required to compute the Mth output, we can do the computation at f_s/M (ie lower computation rate)

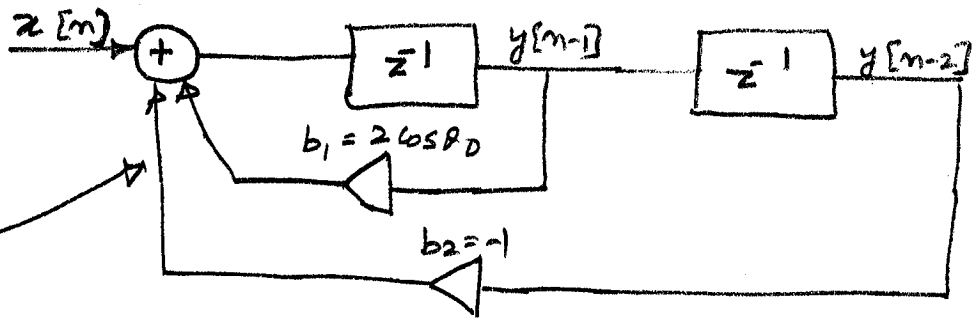
(ii) The cut off frequency (min.) of the lowpass filter must be 3 kHz (because the input signal is band limited to 3 kHz)

(b)





(c) $y[n] = 2r\cos\theta_0\, y[n-1] - r^2 y[n-2] + x[n] - r\cos\theta_0\, x[n-1]$



let $x[n] = 0$, $y[n] = 2r\cos\theta_0\, y[n-1] - r^2 y[n-2]$

For an oscillator $r = 1$ (Poles on the unit circle)

$\therefore y[n] = 2\cos\theta_0\, y[n-1] - y[n-2]$

output: $y[n] = \sin(\theta_0 n)$ ← given

$\therefore y[0] = \sin(0) = 0$; $y[-1] = \sin(-\theta_0) = -\sin\theta_0$

$$\boxed{\begin{array}{l} y[0] = 0 \\ y[-1] = -\sin\theta_0 \end{array}}$$ ← initial conditions

End of TLT - level 2.

TLT - Level 3

(i) $H_d(\theta) = \begin{cases} e^{-j3\theta} & 0 \le |\theta| \le \pi/2 \\ 0 & \text{elsewhere} \end{cases}$

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{-j3\theta} \cdot e^{jn\theta} \, d\theta = \frac{1}{2\pi} \left[\frac{e^{j(n-3)\theta}}{(n-3)j} \right]_{-\pi/2}^{\pi/2}$$

$$= \frac{1}{2\pi j (n-3)} \left(e^{j(n-3)\pi/2} - e^{-j(n-3)\pi/2} \right)$$

$$= \frac{1}{\pi (n-3)} \left(\frac{e^{j(n-3)\pi/2} - e^{-j(n-3)\pi/2}}{2j} \right)$$

$$= \sin(n-3)\frac{\pi}{2}$$

$$\boxed{h_d[n] = \frac{\sin\left[(n-3)\frac{\pi}{2}\right]}{\pi (n-3)}}$$

(ii) $\beta = \frac{N-1}{2} = \frac{7-1}{2} = 3$

For a linear phase filter, symmetrical coefficients centred around 3

$h_d[3] = \frac{1}{2}$ {for $n=3$, use $\sin\theta \approx \theta$ when θ is small or $\lim_{\theta \to 0} \frac{\sin\theta}{\theta} \to 1$}

$h_d[4] = \frac{1}{\pi}$

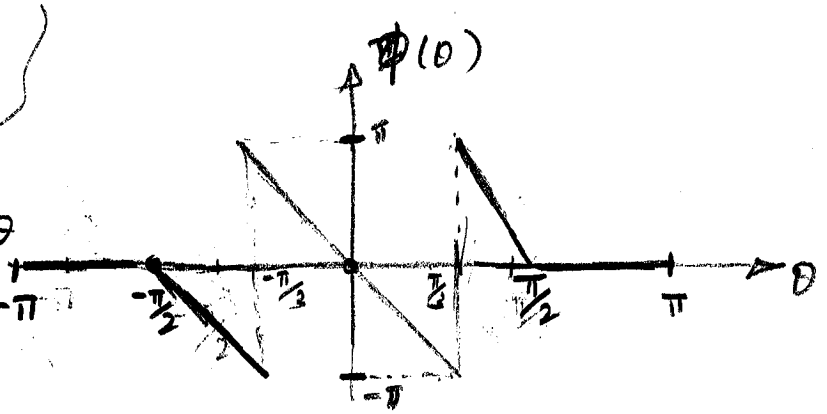
$h_d[5] = 0$

$h_d[6] = \frac{-1}{3\pi}$

n	0	1	2	3	4	5	6
$h_d[n]$	$-\frac{1}{3\pi}$	0	$\frac{1}{\pi}$	$\frac{1}{3}$	$\frac{1}{\pi}$	0	$-\frac{1}{3\pi}$

(iii) $H_d(\theta) = \begin{cases} e^{-j3\theta} \\ 0 \end{cases}$

$\phi(\theta) = -3\theta$



(b) $$G(s) = \frac{1}{s^2+s+1}$$

We must prewarp the analogue cut-off frequency ω_c

$$\omega_c' = \frac{2}{T}\left(\tan\frac{\theta}{2}\right)$$

ω_c' ← new analogue frequency

$\theta = 2\pi \frac{f_a}{f_s}$

$= 2\pi \frac{2\ kHz}{8\ kHz}$

$$\omega_c' = \frac{2}{T}\tan\left(2\pi\frac{2}{8}\right) = \frac{2}{T}$$

$G(s) = \frac{1}{s^2+s+1}$, ∴ The denormalised analogue transfer function is

$$G_1(s) = G(s)\Big|_{s=\frac{s}{\omega_c'}} = \frac{1}{\left(\frac{s}{\omega_c'}\right)^2+\frac{s}{\omega_c'}+1}$$

$$G_1(s) = \frac{(\omega_c')^2}{s^2+\omega_c' s+(\omega_c')^2}$$

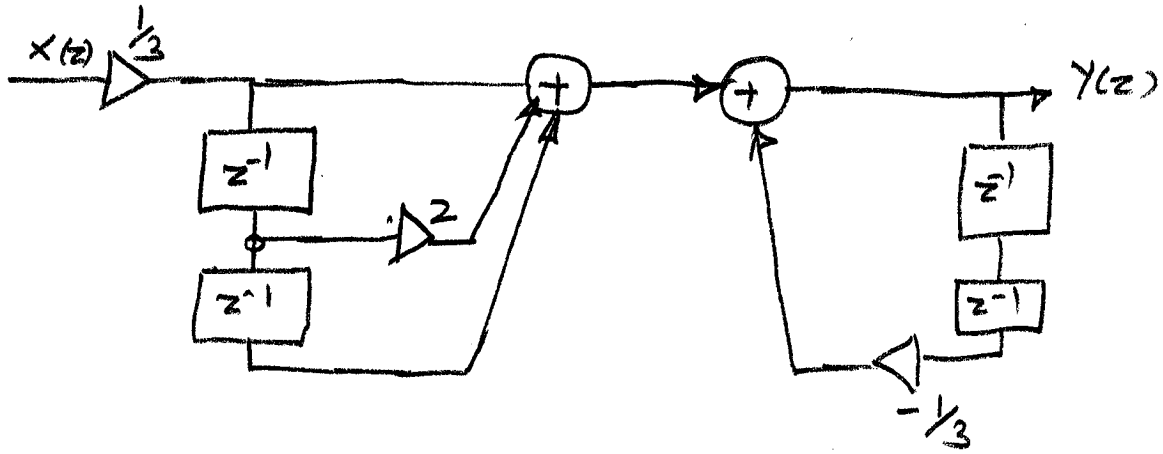
$$\therefore G(z) = G_1(s)\Big|_{s=\frac{2}{T}\frac{z-1}{z+1}} = \frac{\left(\frac{2}{T}\right)^2}{\left(\frac{2}{T}\frac{z-1}{z+1}\right)^2+\frac{2}{T}\cdot\frac{2}{T}\frac{z-1}{z+1}+\left(\frac{2}{T}\right)^2}$$

$$= \frac{1}{\frac{(z-1)^2}{(z+1)^2}+\frac{z-1}{z+1}+1}$$

$$G(z) = \frac{(z+1)^2}{z^2-2z+1+z-1+1} = \frac{(z+1)^2}{3z^2+1}$$

$$\boxed{G_1(z) = \frac{(z+1)^2}{3\left(z^2+\frac{1}{3}\right)}}$$

$$G(z) = \frac{1}{3}\,\frac{z^2+2z+1}{z^2+\frac{1}{3}} = \frac{1}{3}\cdot 1+2z^{-1}+z^{-2} \circ \frac{1}{1+\frac{1}{3}z^{-2}}$$



TLT - Level 4

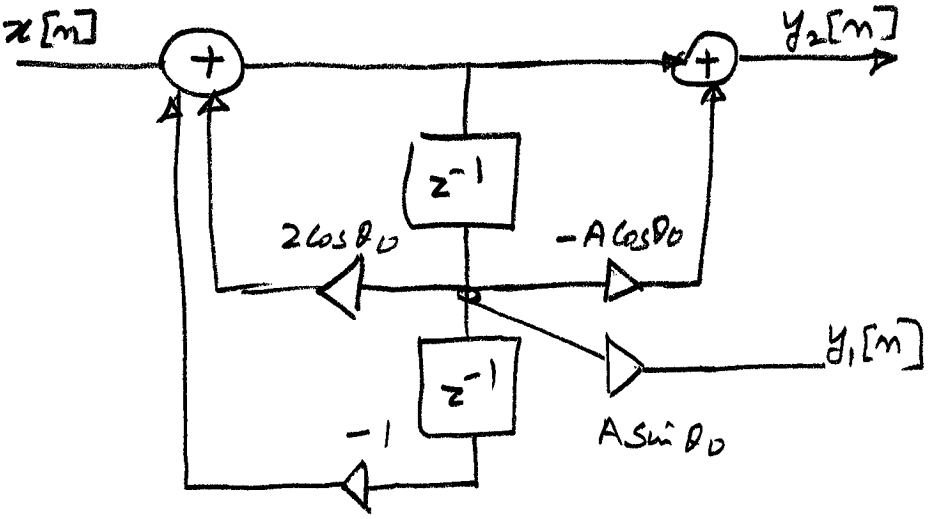
(a) (i) $h_1[n] = A\sin(n\theta_0)\,u[n]$, $\therefore H(z) = A\,\dfrac{z^{-1}\sin\theta_0}{1-2\cos\theta_0 z^{-1}+z^{-2}}$

Hence $y_1[n] = y[n] = 2\cos\theta_0\, y[n-1] - y[n-2] + A\sin\theta_0\, x[n-1]$ — (1)

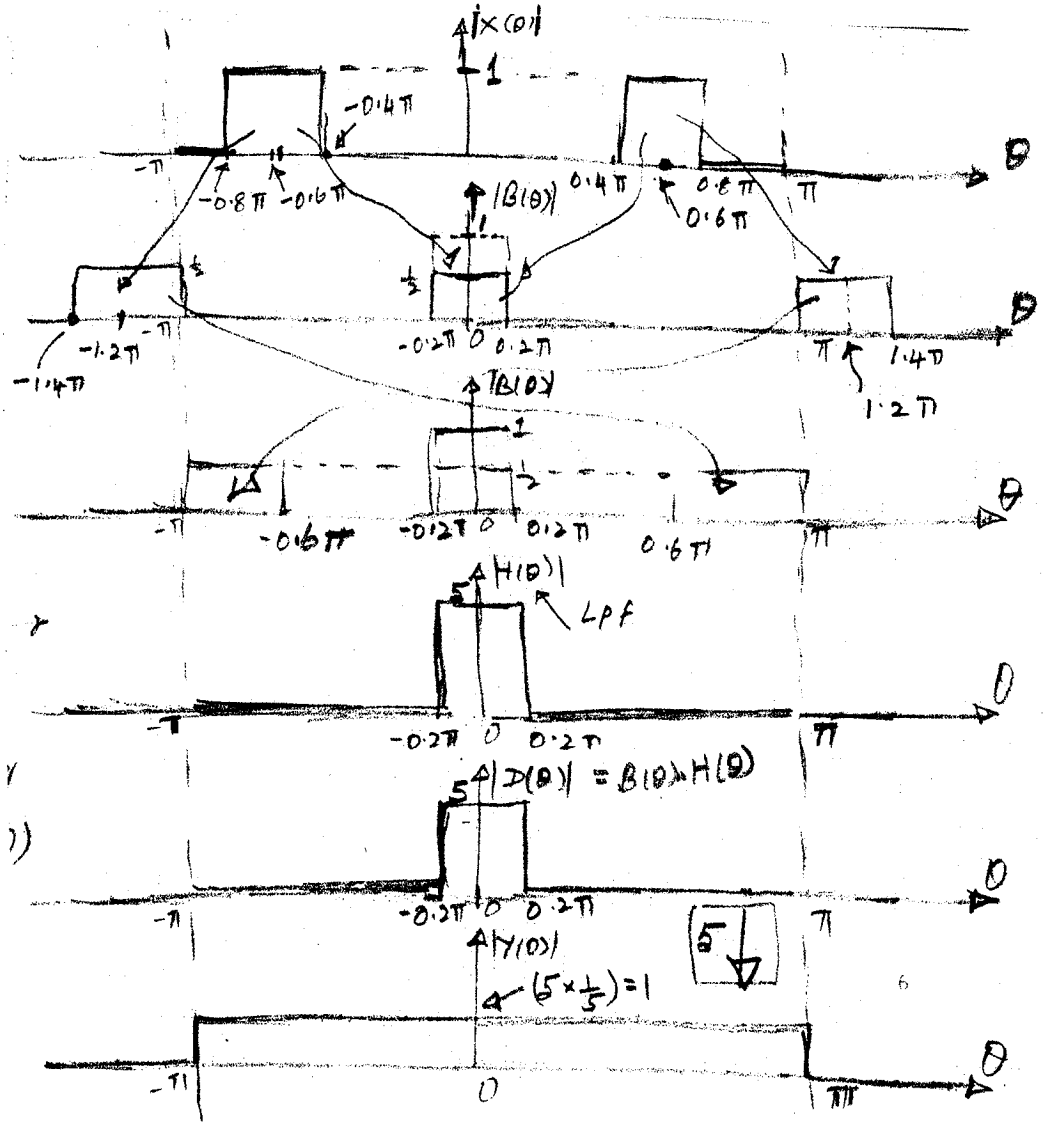
$h_2[n] = A\cos(n\theta_0)\,u[n]$, $\therefore H(z) = \dfrac{A\,(1-z^{-1}\cos\theta_0)}{1-2\cos\theta_0 z^{-1}+z^{-2}}$

Hence $y_2[n] = y[n] = 2\cos\theta_0\, y[n-1] - y[n-2] + A\,x[n] - A\cos\theta_0\, x[n-1]$ — (2)

(ii)



(b)



TLT- Level 5

(a) $\theta_1 = \frac{\omega_1 T}{2} = \omega_1 \times \frac{1}{2f_s} = 2\pi \times 2\times10^3 \times \frac{1}{2\times8\times10^3} = \frac{\pi}{4}$

$\theta_2 = (\omega_1+\omega_c)\frac{T}{2} = \frac{\omega_1+\omega_c}{2f_s} = \frac{2\pi\times2\times10^3 + 2\pi\times4\times10^3}{2\times8\times10^3} = \frac{3\pi}{4}$

(i) given $X(\theta) = 1+\cos 2\theta$

$b_c(\theta) = \frac{1}{2}\left[X(\theta-\theta_1) + X(\theta+\theta_1)\right] = \frac{1}{2}\left[X(\theta-\frac{\pi}{4}) + X(\theta+\frac{\pi}{4})\right]$

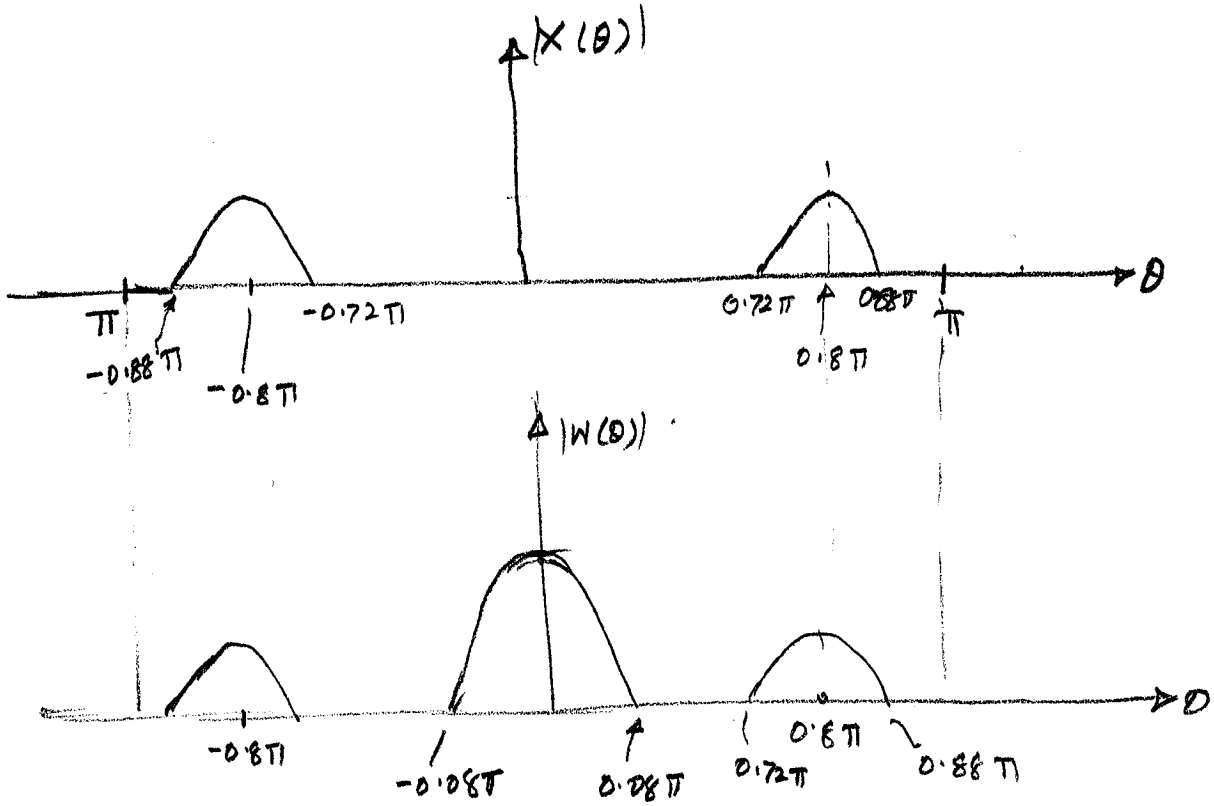
$= \frac{1}{2}\left[\cos\left(2(\theta-\frac{\pi}{4})\right) + \cos\left(2(\theta+\frac{\pi}{4})\right)\right] = \frac{1}{2}\left[1+\sin 2\theta + 1 - 2\sin 2\theta\right]$

$= 1$

(b) $f_L = 900$ $\quad \theta_L = 2\pi \frac{900}{2500} = 0.72\pi$

$f_s = 2500$ Hz

$f_H = 1100$ Hz $\quad \theta_H = 2\pi \frac{1100}{2500} = 0.88\pi$



The ideal lowpass filter $H_d(\theta)$ has a passband of $-0.08\pi \le \theta \le 0.08\pi$, Hence

$$\therefore H_d(\theta) = \begin{cases} 1e^{-j3\theta} & -0.08\pi \le \theta \le 0.08\pi \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore h_d(n) = \frac{1}{2\pi} \int_{-0.08\pi}^{0.08\pi} e^{-j3\theta} \cdot e^{j\theta n} \, d\theta$$

$$= \frac{\sin(0.08\pi(n-3))}{\pi(n-3)}$$

$h_w[n] = h_d[n] \cdot w[n]$ ← rectangular filter

$M = 7$

$\beta = \frac{7-1}{2} = 3$

For linear Phase FIR filter coefficients are symmetrical

n	0	1	2	3	4	5	6
$h_w[n]$				0.08	$\frac{\sin(0.08\pi)}{\pi}$		